

Participation 6

Sections 2.7, 2.8, and 3.1

Due: October 5, 2026

Name: _____

Print out the following pages and write your answers on them. You must show your work. Correct answers without explanation will not receive full credit.

1. A television camera is positioned 8000ft from the base of a rocket launching pad. The angle of elevation of the camera has to change at the correct rate in order to keep the rocket in sight. Let's assume the rocket rises vertically and its speed is 600ft/s when it has risen 3000ft

- (a) Fill in the following table as you progress through each part.

Quantity	Meaning	Value at this instant	Rate of change
x			
y			
c			
θ			

- (b) Draw a diagram representing the given scenario and label with the given quantities in part a.

- (c) Which quantities are constant? How do you know?

- (d) Write an equation that shows how the lengths of the three sides are related.

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- (e) Use your equation in part d and the information in part a to determine how fast the distance from the camera to the rocket changing at that instant.
- (f) Write an equation that shows how the angle and two of the sides are related.
- (g) Use your equation in part f and the information in a to answer the following. If the television is always aimed at the rocket, how fast is the camera's angle of elevation changing at the same moment?

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2. A boat is pulled into a dock by a rope attached to the bow of the boat and passing through a pulley on the dock that is 1 m higher than the bow of the boat. If the rope is pulled in at a rate of 1 m/s , how fast is the boat approaching the dock when it is 5 m from the dock. Use the same strategy from question 1 to answer the question.

Quantity	Meaning	Value at this instant	Rate of change

3. (a) Write down the equation for linearization.

(b) Given the following value $\sqrt[4]{80.9}$, write a function that can be used to approximate this value.?

(c) Choose a number close to 80.9 for which the function value is easy to calculate.

(d) Construct the linearization of your function centered around the value you chose in part c.

(e) Use $L(x)$ to approximate $\sqrt[4]{80.9}$.

4. Explain what a differential is.

5. Find the differential of the following functions.

(a) $y = \frac{s}{6 + 4s}$

(b) $y = x \sin(x)$

6. The edge of a cube was found to be 20 cm with a possible error in measurement of 0.2 cm.

- (a) Describe what the variables are.

- (b) The volume of a cube is $V = x^3$, Use differentials to estimate the maximum possible error in computing the volume of the cube.

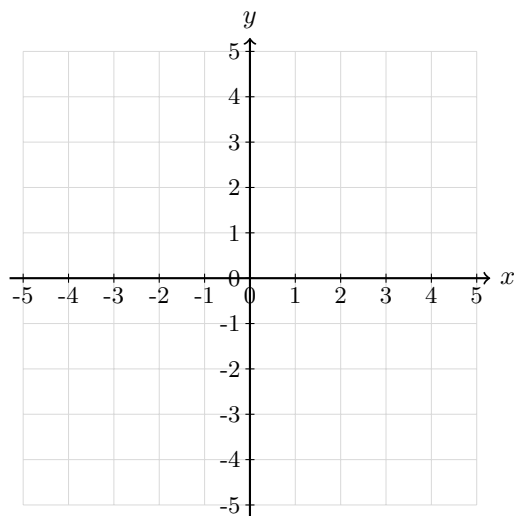
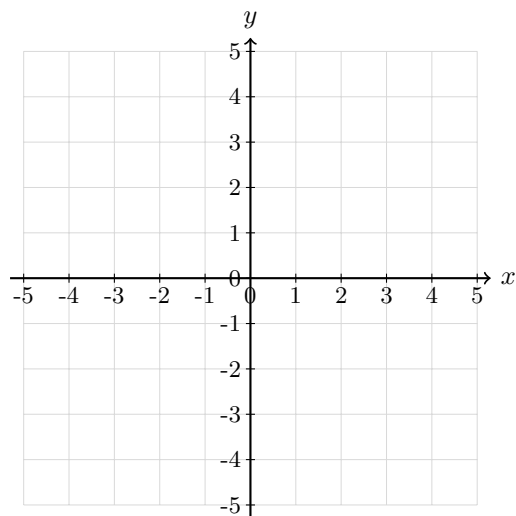
- (c) The relative error can be described as $\frac{\text{absolute error}}{\text{actual value}}$. Find the relative error of the volume of the cube.

- (d) What is the percentage error of the volume of the cube.

- (e) The surface area of a cube is $S = 6x^2$. Use differentials to estimate the maximum possible error, relative error, and percentage error in computing the surface area of the cube.

7. Let $f(x) = e^x$ and $g(x) = e^{-x}$

(a) Graph $f(x)$ and $g(x)$. Label as accurately as possible.



Using your graphs in part (a), find the following limits.

(b) $\lim_{x \rightarrow \infty} e^x$

(c) $\lim_{x \rightarrow -\infty} e^x$

(d) $\lim_{x \rightarrow \infty} e^{-x}$

(e) $\lim_{x \rightarrow -\infty} e^{-x}$

(f) $\lim_{x \rightarrow -\infty} e^{x^2}$

(g) $\lim_{x \rightarrow \infty} \frac{e^{7x} + 9e^{2x}}{e^{-5x} + 8e^{7x}}$